

Adilet Shynggys

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RESEARCH INTERESTS

Heat transfer, thermodynamics, and nuclear energy systems, with emphasis on reactor thermal-hydraulics — subcooled boiling, departure from nucleate boiling, and critical heat flux — grounded in interfacial and multiphase fluid mechanics, capillary and contact-line dynamics, and reduced-order modeling of two-phase transport.

RESEARCH EXPERIENCE

Undergraduate Researcher — Capillary-Driven Two-Phase Transport

August 2026 – Present

Department of Mechanical and Aerospace Engineering, Nazarbayev University • Advisor: Prof. Amgad Salama

Designing capillary-driven surface textures to passively transport liquid slugs and droplets along combustion-chamber heat-exchanger pipes in power plants, reducing flow maldistribution and heat loss without external pumping.

- Extending a semi-analytical framework for wetting and non-wetting liquid penetration through cylindrical and tapered capillaries, resolving meniscus velocity, dynamic contact angle, and film thickness, via coupled momentum–capillarity equations formulated and numerically integrated in MATLAB and Python.
- Developing reduced-order models of capillary-driven two-phase transport that capture the dominant interfacial physics at far lower cost than 3D simulation.
- Designing tapered SU-8 microchannels for photolithographic fabrication on silicon to test the model experimentally: the width gradient creates a Laplace-pressure asymmetry driving wetting liquids toward the narrow end and non-wetting liquids toward the wide end.
- Connecting this geometry-driven transport to passive liquid return in two-phase thermal management and to rewetting in heated channels.

Undergraduate Researcher — Triboelectric Energy Conversion

Jan. – Dec. 2025

Energy Conversion Laboratory, Nazarbayev University

- Fabricated triboelectric nanogenerators for self-powered wearable sensing (e.g., robotic-hand and haptic feedback systems, hiking/outdoor gear) via electrospinning and silicone casting; varied surface morphology and material pairing to raise output power density by 15–20%.
- Built the electrical characterization workflow (open-circuit voltage, short-circuit current, load sweeps) and processed the resulting datasets in OriginPro and MATLAB.
- Contributed experimental data and analysis to a co-authored article in *Small* and to an award-winning conference poster.

EDUCATION

Nazarbayev University

B.Eng. in Mechanical and Aerospace Engineering

Astana, Kazakhstan

Expected June 2027

- GPA: 3.86 / 4.00 — ranked 1st in cohort; Dean’s List, four consecutive semesters.
- Relevant coursework: Fluid Mechanics; Engineering Thermodynamics; Numerical Methods; Differential Equations and Linear Algebra; Engineering Dynamics I–II; Structural Mechanics; Materials and Manufacturing; CAD. In progress: Heat and Mass Transfer; Mechanical Systems Design; Aerospace Propulsion.

The University of Hong Kong

Visiting Exchange Student, Department of Mechanical Engineering

Hong Kong SAR

Spring 2026

- Coursework in Energy Conversion Systems, Design and Manufacture, and Additive Manufacturing; completed two design projects during the exchange (see Engineering Design Projects) and collaborated with Prof. Wei Yan on curriculum design for an undergraduate course.

PUBLICATIONS AND CONFERENCE PRESENTATIONS

- [1] S. Faramarzi, S. Yavari, U. Ospanova, **A. Shynggys**, et al. “Polarity-Engineered, Tribo-Positive Imidazolium-Mediated Crosslinked 6FDA-Durene Films for Triboelectric Nanogenerator,” *Small*, 2026, art. e74486. <https://doi.org/10.1002/smll.74486>
- [2] **A. Shynggys**, S. Yavari, G. Kalimuldina. “Green-Synthesized Ag Nanoparticles Incorporated into High-Performance Triboelectric Wearable Sensors.” Presented as a poster at the Belt and Road Initiative Conference, Astana, 2025 (**Best Research Poster Award**), and as an abstract at the 13th International Conference on Nanomaterials and Advanced Energy Storage Systems (INESS-2025), Nazarbayev University, Astana, Aug. 2025.

ENGINEERING DESIGN PROJECTS

Reduced-Order Design and Analysis of a VVER-1200 Reactor Core

August 2026 – Present

Senior Capstone Project, Nazarbayev University • Team of 4 students, 1 supervisor • In progress

- Leading a four-student team designing a VVER-1200 core from scratch: from required electric power, to thermal power via cycle efficiency, to fuel element/assembly geometry and hexagonal core configuration.
- Developing a physics-based reduced-order model coupling simplified neutronic and thermal-hydraulic sub-models — multiplication factor, power distribution, coolant temperature, pressure drop, heat transfer, and margins to subcooled boiling and departure from nucleate boiling — each feeding the other.
- Verifying and validating the model against published reference data and benchmark problems, enabling rapid parametric design studies and preliminary safety assessment.

Other Engineering Design Projects

2025 – Spring 2026

- *Ergonomic Train-Driver Seat Design* (Mechatronics Engineering Case Competition, sponsored by Wabtec Corporation; team of 3, 1st Place): designed and structurally analyzed a driver's seat for 14-hour locomotive shifts, certifying it against GOST 33330-2015 under static/dynamic fatigue loading.
- *Multi-Stage Transmission (Gearbox) Design* (University of Hong Kong, Prof. Wei Yan): sized shafts using the ASME combined bending-torsion criterion and selected gear modules, keyways, and bearings to meet the project's specified safety margin and fatigue life.
- *Interactive Public Art Installation* (University of Hong Kong, Prof. Homayoon Kazerooni, UC Berkeley): modeled flower-petal forms in SolidWorks to specified dimensional requirements, arranging multiple petals to avoid geometric collisions and verifying structural stability, then 3D-printed each piece on a Bambu Lab A1.

INDUSTRY EXPERIENCE

Product Support Engineering Intern

June – July 2025

Wabtec Corporation — Kazakhstan

- Analyzed thermal performance of locomotive traction motors and supported root-cause review of field temperature excursions.
- Audited shop-floor manufacturing against engineering drawings with a fellow intern, reconciling GOST (shop) with ASME (Wabtec's U.S. standard).
- Traced an insulation-usage inconsistency across units to climate differences between export markets (e.g., uninsulated units for India vs. insulated units for Kazakhstan's winters).

AWARDS AND HONORS

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| • Presidential Scholarship of the Republic of Kazakhstan — national merit scholarship | 2026–2027 |
| • NU President's Scholarship — full-tuition university merit award | 2024–2025 |
| • Yessenov Foundation Scholarship — top 20 of 500+ applicants | 2026–2027 |
| • Best Research Poster, Belt and Road Initiative Conference | 2025 |
| • 1st Place, Mechatronics Engineering Case Competition (sponsored by Wabtec) | 2025 |
| • Champion, NU AIChE Engineering Challenges | 2025 |

TECHNICAL SKILLS

Programming	MATLAB, Python, Mathematica, $\LaTeX$
Modeling & Numerics	Reduced-order modeling of multiphase transport; nonlinear ODE integration; validation against analytical limits
Simulation & CAD	SolidWorks (CSWA certified), OriginPro
Experimental	Electrospinning, silicone casting, 3D printing, electrical device characterization
Design Codes	ASME shaft design, GOST 33330-2015
Languages	Kazakh (native), Russian (fluent), English (IELTS Overall Band 8.0)

TEACHING, LEADERSHIP, AND SERVICE

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| • Physics Tutor, Nazarbayev University — 20+ first-year students. | 2023–2024 |
| • Cohort Representative, Mechanical and Aerospace Engineering. | 2024–2026 |
| • Organizer, Charity Trail Run — 150 runners, 600,000 KZT (~\$1,300) raised. | 2025 |
| • Volunteer, HKU Muslim Society and Mereke Charity Foundation. | 2025–2026 |